ECON 1550: International Finance

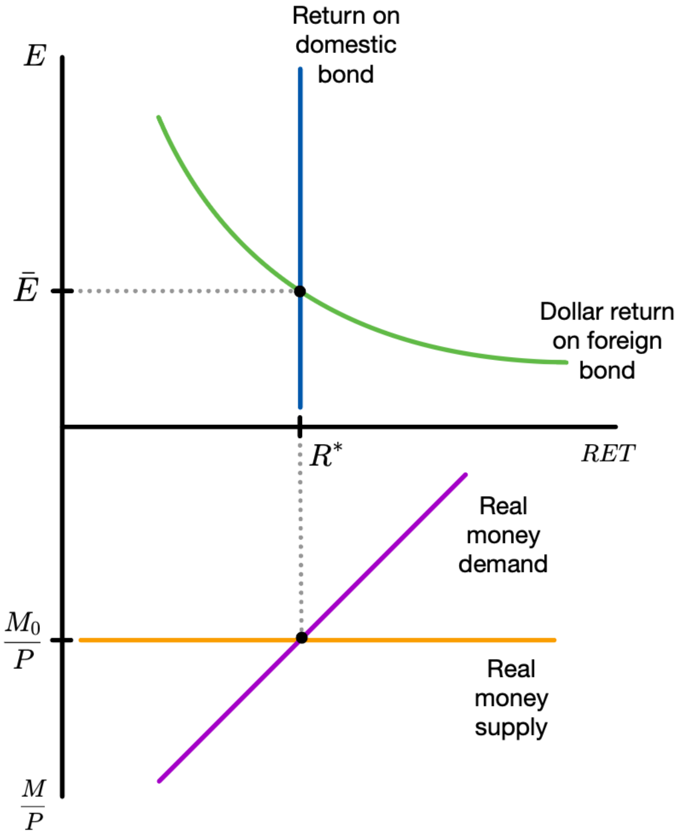
Fixed Exchange Rates and Foreign Exchange Intervention

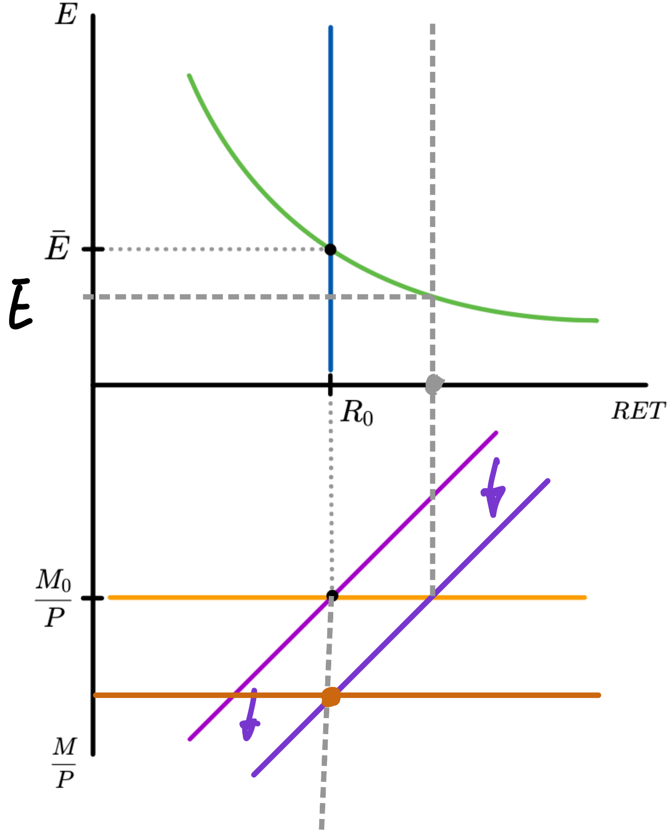
How Central Banks Fix Exchange Rates

- Want to have a fixed exchange rate $E = \bar{E}$
- Assume $E = \bar{E}$ forever: the fixed exchange rate regime is said to be credible
- Then $E^e = E = \bar{E}$
- Central bank changes money supply so that $E = \bar{E}$ at all times
- UIP with $E^e = E$ implies $R = R^*$
- In a fixed exchange rate regime, a depreciation is called a devaluation and an appreciation a revaluation

Y ↑ Y* UNCHAN.

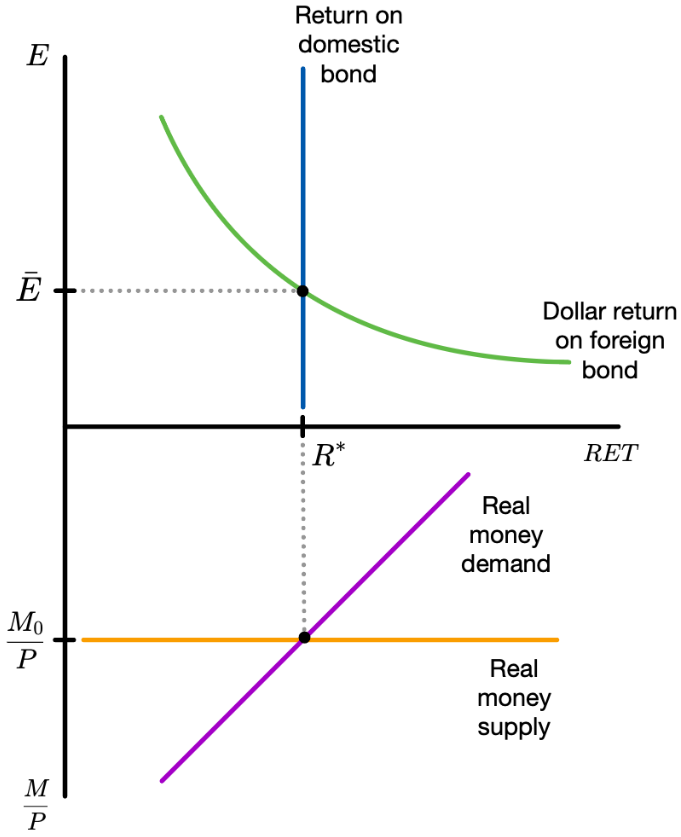
Response to Changes in Y

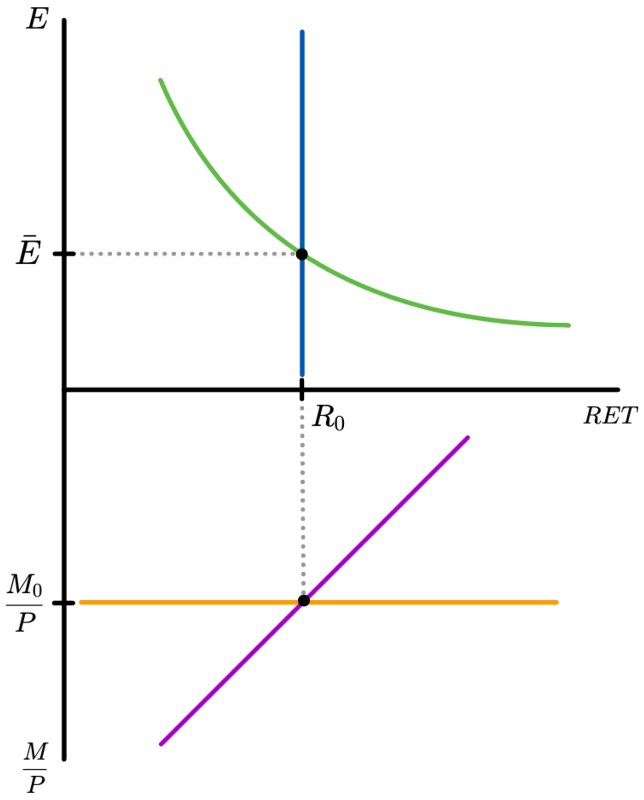




Response to Changes in R^*

LEFT AS EXERCISE :)

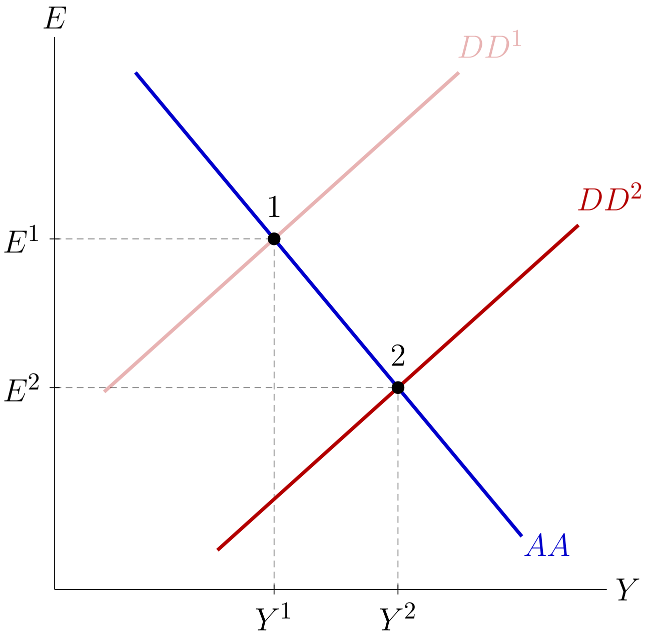




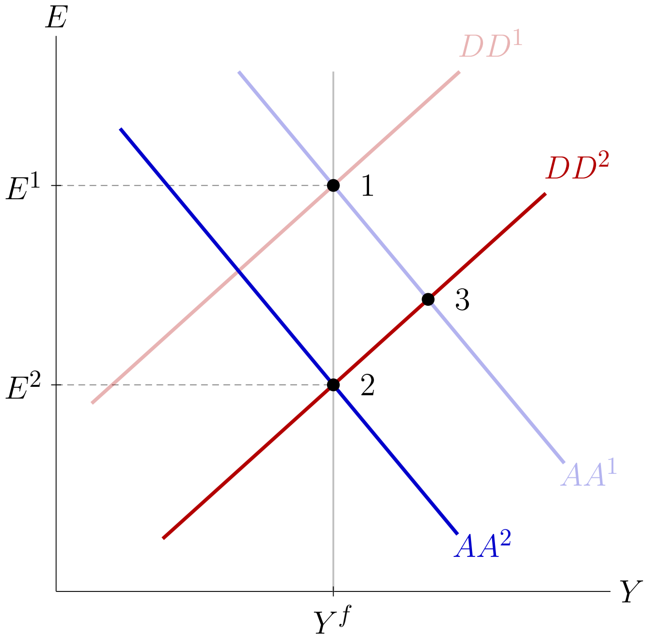
Fiscal Stimulus

Floating exchange rate

Temporary

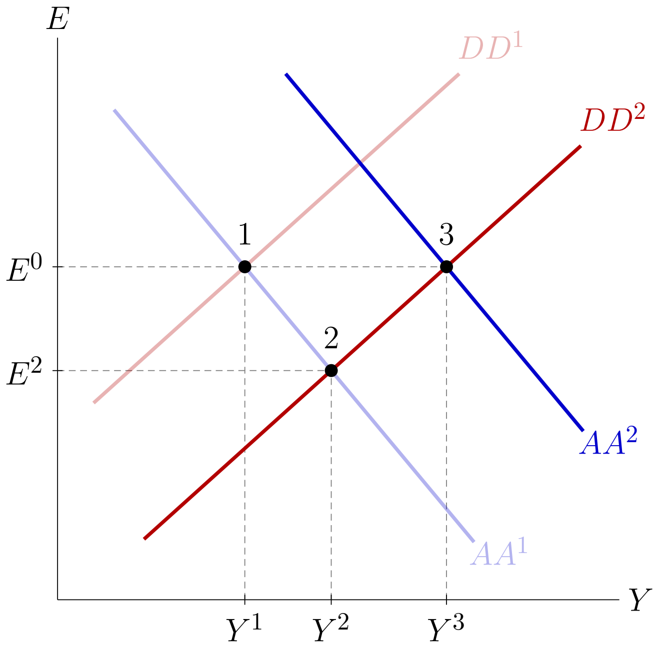


Permanent



Fixed exchange rate

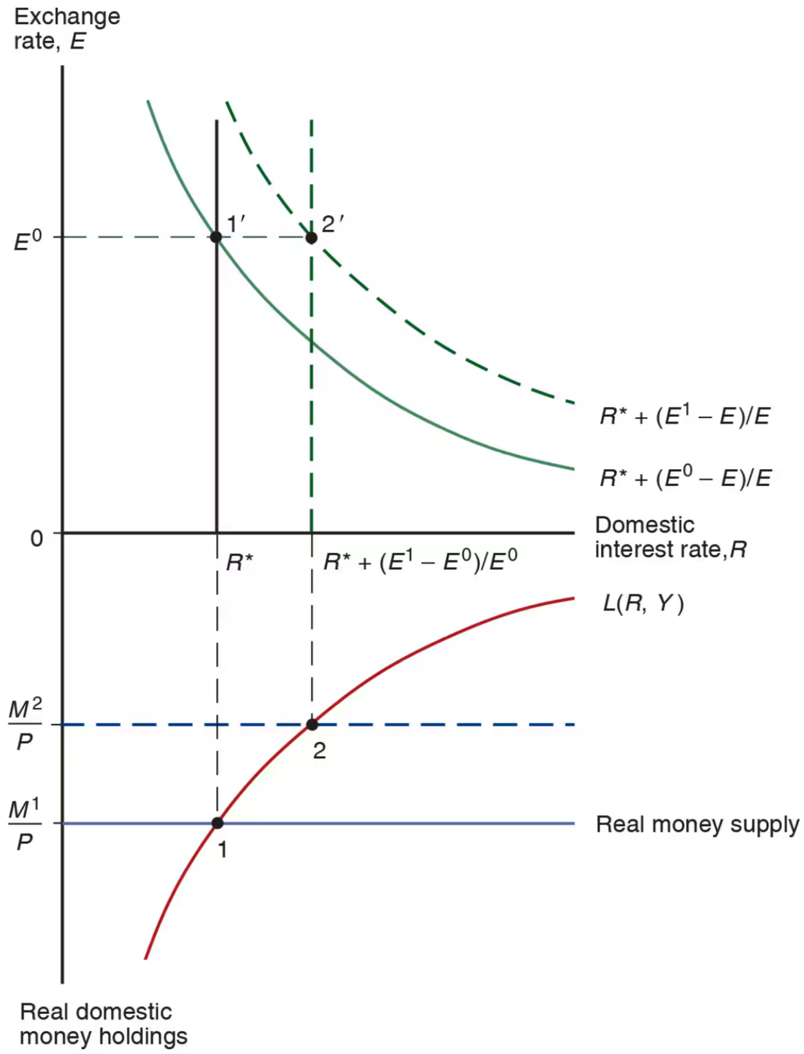
Temporary or permanent



Balance of Payments Crises

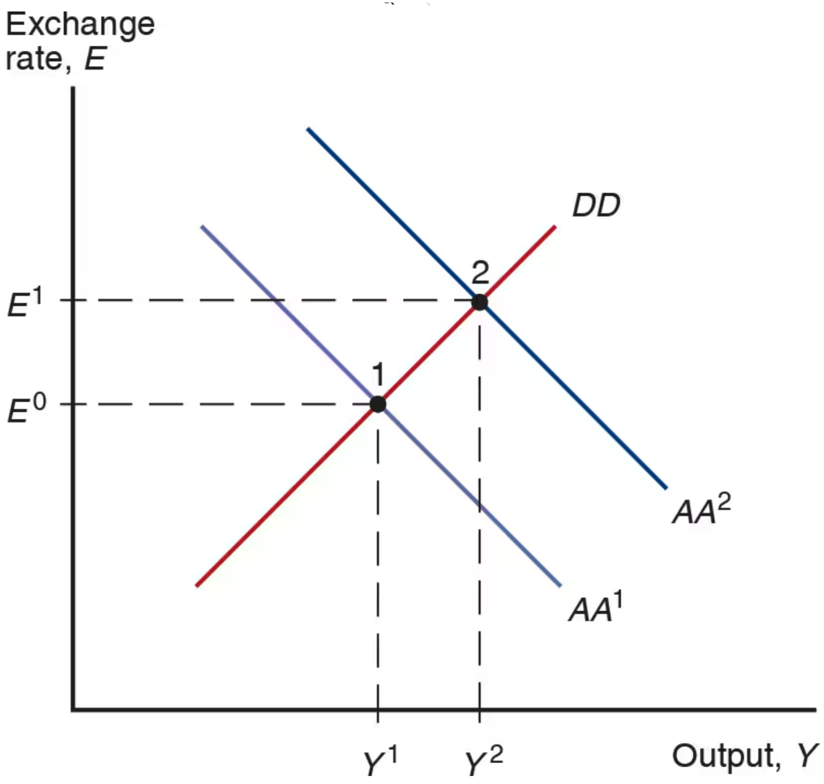
- Want to have a fixed exchange rate $E = \bar{E}$
- If $E^e = E = \bar{E}$ forever, the fixed exchange rate regime is credible
- Imagine E^e suddenly increases
- Without any government intervention, $E^e < \bar{E}$

- To maintain $E = \bar{E}$, money supply must decline from M^1 to M^2
- The domestic interest rate must increase to $R^* + E^1/E^0 - 1$
- As money supply declines, R is always below the dollar return on the foreign bond
- Investors flee the domestic currency and domestic bond: a **capital flight**



Devaluation

- If central bank runs out reserves, or expects it will run out in the future, there will be a devaluation
- Fixed exchange rate must be modified or abandoned



1 PESO = 1 DOLLAR

FULLY CREDIBLE $E = \bar{E}$? INSTITUTIONS MISTRUST.

END: DEVALUATION 3 PESOS = 1 USD

(1) 1 PESO = 1 USD → APPRECIATION

(2) EX $\downarrow$ IM $\uparrow$ $Y \downarrow$ $u \uparrow$

($Y \downarrow$ PHILLIPS CURVE $\pi \downarrow$)

(3) $G \uparrow$ DEFICITS GO UP

(4) TEQUILA & ASIAN CRISES

(5) BRAZIL (BIGGEST TRADING PARTNER) DEVALUES EX ↓

(6) G ↑ BECAUSE Y ↓ u ↑

(7) HIGH GOV DEBT SOLVED BY:

- ABANDON $E = \bar{E}$
- DEFAULT
- HIGH π → ABANDON $E = \bar{E}$

(8) RIOTS!

(9) GOV. COLLAPSE

- UNSTABLE GOV.
- DEVALUATION
- DEFAULT

(10) NEXT 3 YEARS WERE AMAZING

- NO DEBT SERVICE
- SUPER COMPETITIVE EX
- GRAIN PRICES WORLDWIDE

WENT UP.

⋮

(N) 10 YEARS LATER: DEFAULT AGAIN